

Input vs. Output	• What are users inputting? • What are the outputs or labels your model will predict or generate?	Input: A natural-language question typed by a smallholder farmer or extension worker (text), e.g. "Why are my maize leaves turning yellow?" There is no image, audio, or structured input in this stage of the system. Output: A ranked list of the 5 most relevant document IDs from a fixed knowledge base of 695 agricultural-extension factsheets, ordered best-first. This is the retrieval component of a larger RAG pipeline, the output here is <i>which documents to read</i>, not a generated natural-language answer. In the full system, these top-5 documents would subsequently be passed to an LLM to compose the final farmer-facing answer, but that generation step is out of scope for this retrieval benchmark.
Classification problem	• What is the classification task, if any? • What categories/classes are you working with, and why?	Strictly, this is a ranking/retrieval task rather than classification, the model orders documents by relevance rather than assigning a single label. However, training and evaluation rely on a 4-level graded relevance scale applied to (query, document) pairs: 3 — Perfect: directly answers the question. 2 — Relevant: a genuinely complementary facet of the same topic. 1 — Marginal: the same issue on a different crop. 0 — Not relevant (including deliberate hard negatives, lexically similar documents that answer

		the wrong intent, e.g. "prevention" when the question asks about "causes/management"). These four grades were chosen specifically to separate true semantic/intent understanding from naive keyword matching, which is the central technical challenge in this domain, a farmer asking "how do I manage X" must not be answered with a document about "preventing X," even though the two are nearly identical in wording.
Labeling	• Where do your labels come from? • How will you check for consistency and accuracy?	The graded relevance judgments (qrels_train.csv) were provided as ground truth by the challenge organizers, produced according to the rubric above rather than by our team directly. Our team did not perform first-party annotation for this benchmark dataset. To check consistency and catch labeling issues before trusting the data: we manually inspected the judged documents for individual queries against their titles and content (e.g. verifying that "Managing Bacterial Leaf Blight" variants across agro-zones were all graded 3, while lexically similar "Preventing" variants were correctly graded 0), and we built an independent scoring harness (nDCG@5 against qrels_train) to continuously sanity-check that our retrieval model's behavior is consistent with the intent of the labels, not just their surface wording. If our team extends this dataset with new documents or queries in future iterations, we would define an explicit labeling rubric (matching the 0–3 scale

		above) and have at least two domain-literate reviewers grade a shared sample to measure inter-annotator agreement before scaling up.
Data creation	• How will you collect your data? • Direct sources vs. synthesized data?	The knowledge base (documents.csv, 695 factsheets) is not collected by our team from scratch; each document is tagged with an origin field as either: llm_grounded: rewritten/summarized from real, existing agricultural-extension publications (with a source_url and CC-BY license recorded for traceability), attributed to named institutions such as FAO, CGIAR, Plantwise, IITA, ICRISAT, and AGRA. synthetic: generated to fill coverage gaps (e.g. producing consistent agro-zone variants of the same topic, irrigated lowland, Guinea savanna, humid forest, so the corpus has comparable depth across regions) without a single external source document. In practice the corpus is overwhelmingly synthetic: 637 of 695 documents (91.7%) are synthetic, and only 58 (8.3%) are llm_grounded with a verifiable source_url. The source field (e.g. "CGIAR," "FAO") is populated for all 695 documents, but for the synthetic majority it denotes the institutional <i>voice/style</i> the content was generated to match, not a real external document we can point to. We treat this distinction as important to disclose rather than blur. The queries (train_queries.csv, test_queries.csv) are written in the voice of a farmer's question but are not

		transcriptions of real farmer interviews or field recordings, they simulate realistic phrasing at scale. This is a known limitation for our own project's later stages: real farmer input (especially voice-transcribed, code-mixed, or non-English queries) will differ from this clean, synthetic text and should be validated against before deployment.
Web collection ethics	• Do you plan to source data from the web? • Have you addressed consent, copyright, and privacy?	Our team did not scrape the web directly for this benchmark. The "llm_grounded" portion of the corpus — 58 of 695 documents (8.3%) — is grounded in existing, already-published extension material, explicitly licensed CC-BY by the original publishers, with the source URL and license recorded per document, allowing reuse with attribution rather than requiring new consent to be sought. The remaining 91.7% of documents are synthetic and have no external URL to verify, they were not scraped, but they are also not independently checkable against a real published source. We are flagging this explicitly rather than implying the whole corpus is web-sourced and licensed; only the grounded subset carries that guarantee. No personal, private, or user-generated data (e.g. forum posts, social media, farmer identities) is included in either subset, all content, grounded or synthetic, is styled as institutional extension guidance intended for public use, which minimizes

		privacy risk even where individual documents aren't externally verifiable. If our project later expands the corpus with additional web-sourced material, we would independently verify each new source's license and attribution requirements rather than assume the same CC-BY status applies, and would avoid scraping any farmer-identifiable or community-generated content without explicit consent.
Bias and representati on	• Who might be left out of your dataset? • How will you improve diversity and representation?	We audited the actual distribution in documents.csv rather than relying on assumption. Two concrete gaps stand out: Crop coverage: Maize, Tomato, Rice, and Cassava alone account for 379/695 documents (54.5%). Pearl millet, a staple crop for arid and Sahelian smallholders, has only 10 documents (1.4%), and Sorghum only 36 (5.2%). Farmers growing drought-resilient staple crops are structurally underserved by this corpus. Country coverage: Nigeria, Ghana, Tanzania, and Kenya together account for 403/695 documents (58%), across just 4 of roughly 48 countries in Sub-Saharan Africa. Countries such as Togo (3 docs), Benin (4), and Rwanda (6) are barely represented, and the large majority of Sub-Saharan African countries have no dedicated documents at all, farmers there would receive only

	generically-worded advice, not content tailored to their agro-ecological or regulatory context. The corpus is also English-only and text-only, which excludes farmers whose primary language is a local/regional language or who rely on voice rather than reading, a direct tension with the accessibility/equity goals in our Challenge 1 problem statement, and a particular concern for the women farmers with less access to formal education named there. Mitigations for our project: treat this benchmark as a technical starting point (the retrieval engine) rather than the final farmer-facing dataset; prioritize sourcing additional documents for underrepresented crops (pearl millet, sorghum) and countries before wider rollout; and treat local-language and voice-input support as required extensions, not optional polish.
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Values and community alignment	• How does your dataset design reflect the values from your problem statement? • How will your community benefit, and how will you remain accountable to them?	Transparency: every document carries a source field naming an institution (CGIAR, FAO, Plantwise, ICRISAT, National Extension Service, AGRA, IITA), and for the 8.3% of documents that are <code>llm_grounded</code>, a <code>source_url</code> makes them independently verifiable. We're not overstating this: for the 91.7% synthetic majority, "source" reflects an attributed style rather than a checkable document, and our system will need to surface that distinction to end users (e.g. "AI-generated, in the style of CGIAR guidance" vs. "sourced directly from CGIAR") rather than presenting both the same way — genuine transparency means disclosing this difference, not just always showing a publisher name. Reliability: we evaluate retrieval quality against expert-graded relevance judgments (<code>nDCG@5</code>) rather than trusting raw keyword overlap, specifically because this dataset is designed to expose intent-confusion errors (e.g. retrieving "prevention" advice for a "causes" question) that could otherwise lead to inappropriate real-world recommendations. Accessibility & equity: we recognize this dataset, as-is, does not yet meet the accessibility bar from our problem statement, it is English-only and text-only. We treat that as an explicit gap to close (local-language and voice support) rather than an acceptable limitation, particularly given the added burden this places on women farmers and
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		lower-literacy users identified in our Challenge 1 problem statement. Accountability: because documents are attributed to real institutions (FAO, CGIAR, Plantwise, IITA, ICRISAT, AGRA) rather than generated anonymously, farmers and extension workers can verify claims against the original publisher, and any inaccuracies can be traced and reported back to a real source rather than an opaque model output.
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